



Structural exploration of the coevolution signal in the T-cell receptor's recognition code

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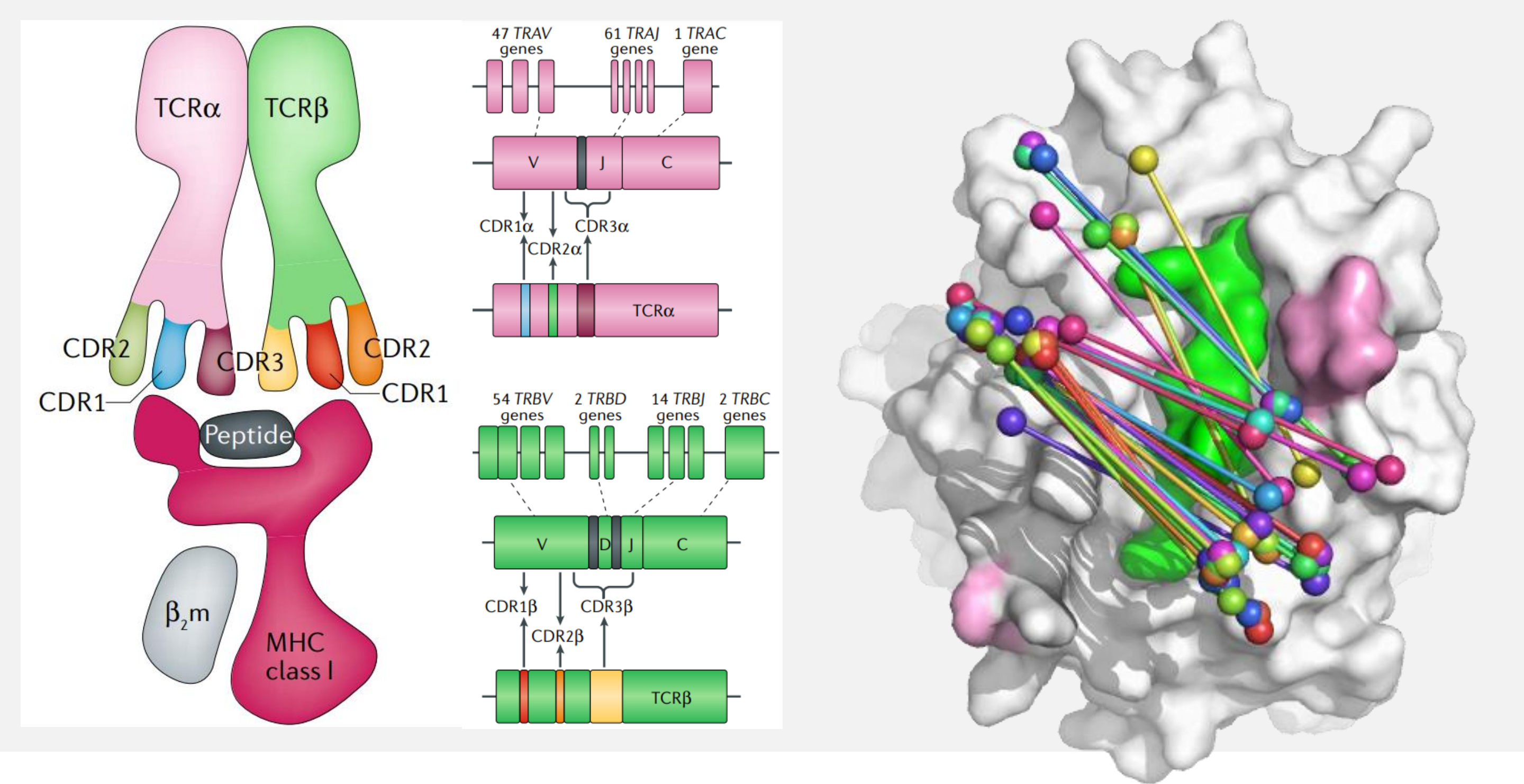
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Summary

The mechanisms behind T-cell receptor (TCR) recognition of peptide-major histocompatibility complex (pMHC) remain not completely understood. Understanding these mechanisms is crucial to drive our knowledge of pathogen and autoimmune responses, as well as for the development of cancer immunotherapies. Germline-encoded motifs in the V-genes of the TCR contribute to their recognition of major histocompatibility complex (MHC) molecules, suggesting co-evolution between TCR binding mechanisms and MHC-exposed residues that shape adaptive immunity. In this work, we investigate the role of coevolution between the TCR and the MHC in TCR-pMHC binding and its impact on AlphaFold predictions.



Approach

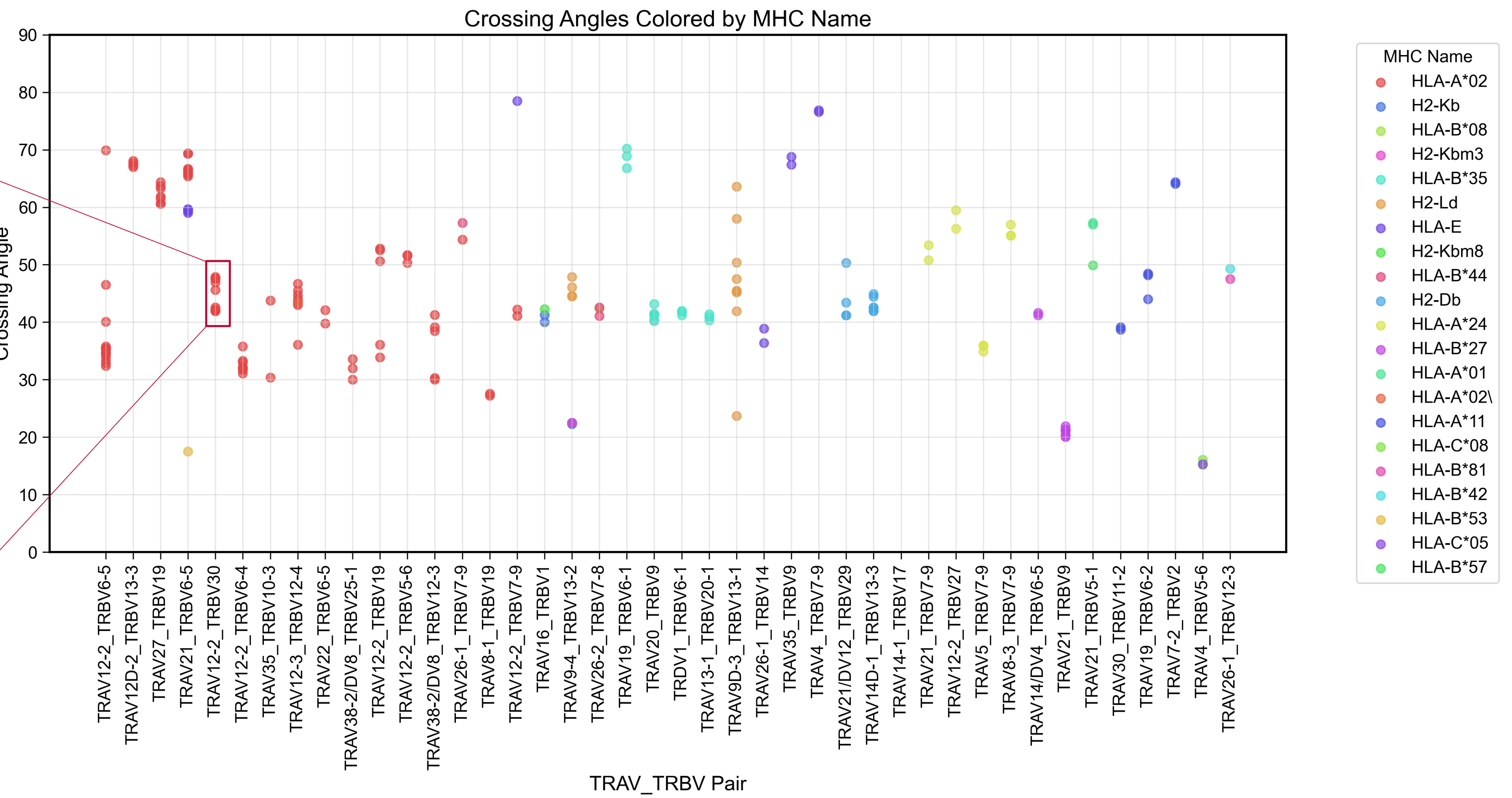
Data: Crossing and Incident angles are calculated from all publicly available MHC Class I pMHC-TCR complexes (downloaded from TCR3d on 11-12-2024)[1]. Previously, it was found that these complexes adopt specific binding orientations, characterized by crossing angles and incident angles, typically ranging from 0° to 90° and 0° to 30°, respectively.

Evolutionary Signal in Binding Orientation:

We found that evolutionarily related structures — those sharing the same TRAV-TRBV combination and MHC allele — tend to adopt similar orientations upon binding, with crossing angles differing by less than 15°. This suggests that coevolutionary relationships shape structural binding patterns.

Hypothesis - Explaining AF3 Performance on Unseen Data:

This observation led to the hypothesis that the presence of strong coevolutionary patterns may explain why AI-based structure prediction models, such as AlphaFold 3 (AF3), perform well better than expected on unseen TCR-pMHC complexes.



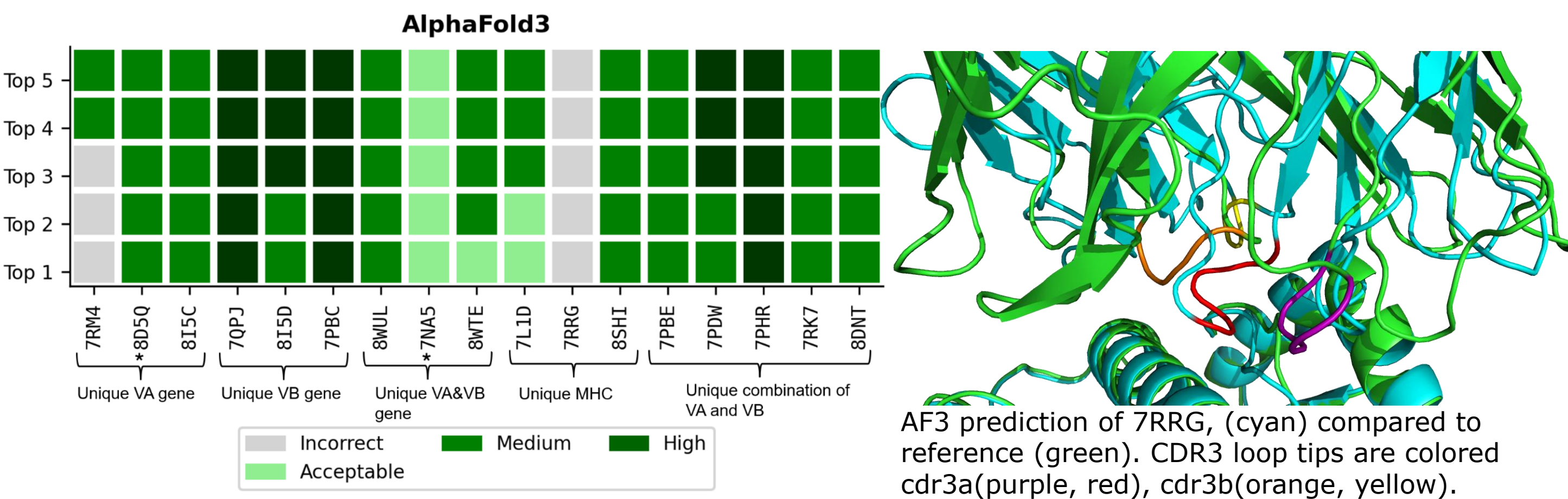
Key Findings

- Structures with identical TRAV and TRBV genes and MHC allele adopt similar orientations upon binding
- Conserved binding orientation can be used to restrict the search space of docking models
- Designed a benchmark set to evaluate AI based models on their dependance of related structures in their available training data

References

- Ragul Gowthaman, Brian G Pierce, TCR3d: The T cell receptor structural repertoire database, Bioinformatics, btz517
- <https://github.com/edsiliav/AF3benchmark>

Construction of benchmark and evaluation of AF3 predictions: We constructed a benchmark where we aimed to limit the coevolutionary patterns while keeping a large variation in TCR-pMHC orientations[2].



AF3 prediction of 7RRG, (cyan) compared to reference (green). CDR3 loop tips are colored cdr3a(purple, red), cdr3b(orange, yellow).



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